

Retail Banking Customer Intelligence & Financial Performance Analysis

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5. Data Quality Assessment
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```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

# Display settings
pd.set_option("display.max_columns", None)

# Plot style
plt.style.use("ggplot")

print("Libraries Imported Successfully!")
```

Libraries Imported Successfully!

```
from google.colab import files

uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving Banking.xlsx to Banking.xlsx

```
clients = pd.read_excel("Banking.xlsx", sheet_name="Clients - Banking")
```

```
all_sheets = pd.read_excel("Banking.xlsx", sheet_name=None)
```

```
clients = all_sheets["Clients - Banking"]
gender = all_sheets["Gender"]
investment_advisor = all_sheets["Investment Advisor"]
banking_relationship = all_sheets["Banking Relationship"]
```

```
import pandas as pd

excel_file = pd.ExcelFile("Banking.xlsx")
print(excel_file.sheet_names)
```

```
['Clients - Banking', 'Gender', 'Banking Relationship', 'Investment Advisor']
```

```
clients.head()
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Feedback
0	IND81288	Raymond Mills	24	34324	2019-05-06	Anthony Torres	American	Safety Technician IV	High
1	IND65833	Julia Spencer	23	42205	2001-12-10	Jonathan Hawkins	African	Software Consultant	High
2	IND47499	Stephen Murray	27	7314	2010-01-25	Anthony Berry	European	Help Desk Operator	High
3	IND72498	Virginia Garza	40	34594	2019-03-28	Steve Diaz	American	Geologist II	Medium
4	IND60181	Melissa Sanders	46	41269	2012-07-20	Shawn Long	American	Assistant Professor	Medium

```
gender.head()
```

	GenderId	Gender
0	1	Male
1	2	Female

```
investment_advisor.head()
```

	IAId	Investment Advisor
0	1	Victor Dean
1	2	Jeremy Porter
2	3	Ernest Knight
3	4	Eric Shaw
4	5	Kevin Kim

```
banking_relationship.head()
```

	BRId	Banking Relationship
0	1	Retail
1	2	Institutional
2	3	Private Bank
3	4	Commercial

```
print("Clients:", clients.shape)
print("Gender:", gender.shape)
print("Investment Advisor:", investment_advisor.shape)
print("Banking Relationship:", banking_relationship.shape)
```

```
Clients: (3000, 25)
Gender: (2, 2)
Investment Advisor: (22, 2)
Banking Relationship: (4, 2)
```

```
df = clients.merge(gender, on="GenderId", how="left")
df = df.merge(banking_relationship, on="BRId", how="left")
df = df.merge(investment_advisor, on="IAId", how="left")
```

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

# Display all columns
pd.set_option("display.max_columns", None)

# Better chart style
plt.style.use("ggplot")

print("Libraries Imported Successfully")
```

```
Libraries Imported Successfully
```

```
clients = pd.read_excel("Banking.xlsx", sheet_name="Clients - Banking")
gender = pd.read_excel("Banking.xlsx", sheet_name="Gender")
investment_advisor = pd.read_excel("Banking.xlsx", sheet_name="Investment Advisor")
banking_relationship = pd.read_excel("Banking.xlsx", sheet_name="Banking Relationship")
```

```
df = clients.merge(gender, on="GenderId", how="left")

df = df.merge(
    banking_relationship,
    on="BRId",
    how="left"
)

df = df.merge(
    investment_advisor,
    on="IAId",
    how="left"
)
```

```
df.head()
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Fee Structure
0	IND81288	Raymond Mills	24	34324	2019-05-06	Anthony Torres	American	Safety Technician IV	High
1	IND65833	Julia Spencer	23	42205	2001-12-10	Jonathan Hawkins	African	Software Consultant	High
2	IND47499	Stephen Murray	27	7314	2010-01-25	Anthony Berry	European	Help Desk Operator	High
3	IND72498	Virginia Garza	40	34594	2019-03-28	Steve Diaz	American	Geologist II	Medium
4	IND60181	Melissa Sanders	46	41269	2012-07-20	Shawn Long	American	Assistant Professor	Medium

```
print("Rows :", df.shape[0])
print("Columns :", df.shape[1])
```

```
Rows : 3000
Columns : 28
```

```
df.columns
```

```
Index(['Client ID', 'Name', 'Age', 'Location ID', 'Joined Bank',
      'Banking Contact', 'Nationality', 'Occupation', 'Fee Structure',
      'Loyalty Classification', 'Estimated Income', 'Superannuation Savings',
      'Amount of Credit Cards', 'Credit Card Balance', 'Bank Loans',
      'Bank Deposits', 'Checking Accounts', 'Saving Accounts',
      'Foreign Currency Account', 'Business Lending', 'Properties Owned',
      'Risk Weighting', 'BRId', 'GenderId', 'IAId', 'Gender',
      'Banking Relationship', 'Investment Advisor'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 28 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Client ID                            3000 non-null   object
1   Name                                3000 non-null   object
2   Age                                  3000 non-null   int64
3   Location ID                          3000 non-null   int64
4   Joined Bank                          3000 non-null   datetime64[ns]
5   Banking Contact                      3000 non-null   object
6   Nationality                          3000 non-null   object
7   Occupation                           3000 non-null   object
8   Fee Structure                        3000 non-null   object
9   Loyalty Classification               3000 non-null   object
10  Estimated Income                     3000 non-null   float64
11  Superannuation Savings               3000 non-null   float64
12  Amount of Credit Cards               3000 non-null   int64
13  Credit Card Balance                  3000 non-null   float64
14  Bank Loans                           3000 non-null   float64
15  Bank Deposits                       3000 non-null   float64
16  Checking Accounts                    3000 non-null   float64
```

```
17 Saving Accounts      3000 non-null float64
18 Foreign Currency Account 3000 non-null float64
19 Business Lending     3000 non-null float64
20 Properties Owned     3000 non-null int64
21 Risk Weighting       3000 non-null int64
22 BRId                 3000 non-null int64
23 GenderId             3000 non-null int64
24 IAIId                3000 non-null int64
25 Gender               3000 non-null object
26 Banking Relationship 3000 non-null object
27 Investment Advisor    3000 non-null object
dtypes: datetime64[ns](1), float64(9), int64(8), object(10)
memory usage: 656.4+ KB
```

The dataset contains customer demographic, banking, financial, and investment information. Most financial columns are numeric and suitable for quantitative analysis

```
df.describe().T
```

	count	mean	min	25%	50%	
Age	3000.0	51.039667	17.0	34.0	51.0	
Location ID	3000.0	21563.323	12.0	10803.5	21129.5	32
Joined Bank	3000	2010-08-15 00:03:21.600000	1995-01-03 00:00:00	2004-04-01 18:00:00	2011-02-18 00:00:00	2017-06:
Estimated Income	3000.0	171305.034094	15919.4754	82906.5943	142313.4796	242290.3
Superannuation Savings	3000.0	25531.599673	1482.03	12513.775	22357.355	354
Amount of Credit Cards	3000.0	1.463667	1.0	1.0	1.0	
Credit Card Balance	3000.0	3176.20678	1.1748	1236.633525	2560.80075	4522.6
Bank Loans	3000.0	591386.155253	0.0	239628.135	479793.4017	825813
Bank Deposits	3000.0	671560.193842	0.0	204400.372299	463316.460618	942754.63
Checking Accounts	3000.0	321092.94907	0.0	119947.526341	242815.659293	434874.93
Saving Accounts	3000.0	232908.353437	0.0	74794.406243	164086.556764	315575.00
Foreign Currency Account	3000.0	29883.529774	44.9984	11916.5464	24341.1872	41966
Business Lending	3000.0	866759.808177	0.0	374825.13645	711314.65755	1185110
Properties Owned	3000.0	1.518667	0.0	1.0	2.0	
Risk Weighting	3000.0	2.249333	1.0	1.0	2.0	
BRId	3000.0	2.559333	1.0	2.0	3.0	
GenderId	3000.0	1.504	1.0	1.0	2.0	
IAId	3000.0	10.425333	1.0	5.0	10.0	

The statistical summary provides the distribution of numerical variables such as income, deposits, loans, savings, and credit card balances.

```
df.isnull().sum()
```

	0
Client ID	0
Name	0
Age	0
Location ID	0
Joined Bank	0
Banking Contact	0
Nationality	0
Occupation	0
Fee Structure	0
Loyalty Classification	0
Estimated Income	0
Superannuation Savings	0
Amount of Credit Cards	0
Credit Card Balance	0
Bank Loans	0
Bank Deposits	0
Checking Accounts	0
Saving Accounts	0
Foreign Currency Account	0
Business Lending	0
Properties Owned	0
Risk Weighting	0
BRId	0
GenderId	0
IAId	0
Gender	0
Banking Relationship	0
Investment Advisor	0

dtype: int64

```
print("Duplicate Rows :", df.duplicated().sum())
```

Duplicate Rows : 0

```
duplicate_clients = df[df["Client ID"].duplicated(keep=False)]
```

```
duplicate_clients
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	St
27	IND35589	Jimmy Simpson	52	38270	2018-01-08	George Lewis	European	Geological Engineer	
53	IND54314	Kevin Gomez	32	24168	2007-01-01	Jerry Green	European	Project Manager	
54	IND74305	Ronald Larson	29	25730	1999-12-11	Bobby Burton	Australian	Software Engineer III	
57	IND26283	Joshua Hughes	58	95	2006-04-21	Shawn Cook	European	Human Resources Assistant I	
71	IND49489	Keith Weaver	74	4567	1995-08-09	Nicholas Cunningham	American	Associate Professor	
...	...	...	...	...	...	...	...	...	...
2948	IND28199	Jesse Wood	70	30301	2002-02-28	Anthony Simpson	European	Staff Accountant IV	
2962	IND85232	Gary Perkins	76	33822	2007-02-12	Jerry Green	European	Geologist I	
2963	IND43081	Gregory Meyer	28	42699	2003-07-05	Raymond Alexander	European	Administrative Assistant IV	
2968	IND80089	Johnny Ryan	79	18443	2009-12-31	James Castillo	Australian	Marketing Assistant	
2976	IND54095	William Alvarez	44	24211	2003-03-23	James Castillo	European	Statistician III	

119 rows × 28 columns

The dataset contains duplicate Client IDs, indicating that the Client ID cannot be treated as a unique primary key.

```
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)
```

df.columns

```
Index(['client_id', 'name', 'age', 'location_id', 'joined_bank',
      'banking_contact', 'nationality', 'occupation', 'fee_structure',
      'loyalty_classification', 'estimated_income', 'superannuation_savings',
      'amount_of_credit_cards', 'credit_card_balance', 'bank_loans',
      'bank_deposits', 'checking_accounts', 'saving_accounts',
      'foreign_currency_account', 'business_lending', 'properties_owned',
      'risk_weighting', 'brid', 'genderid', 'iaid', 'gender',
      'banking_relationship', 'investment_advisor'],
      dtype='object')
```

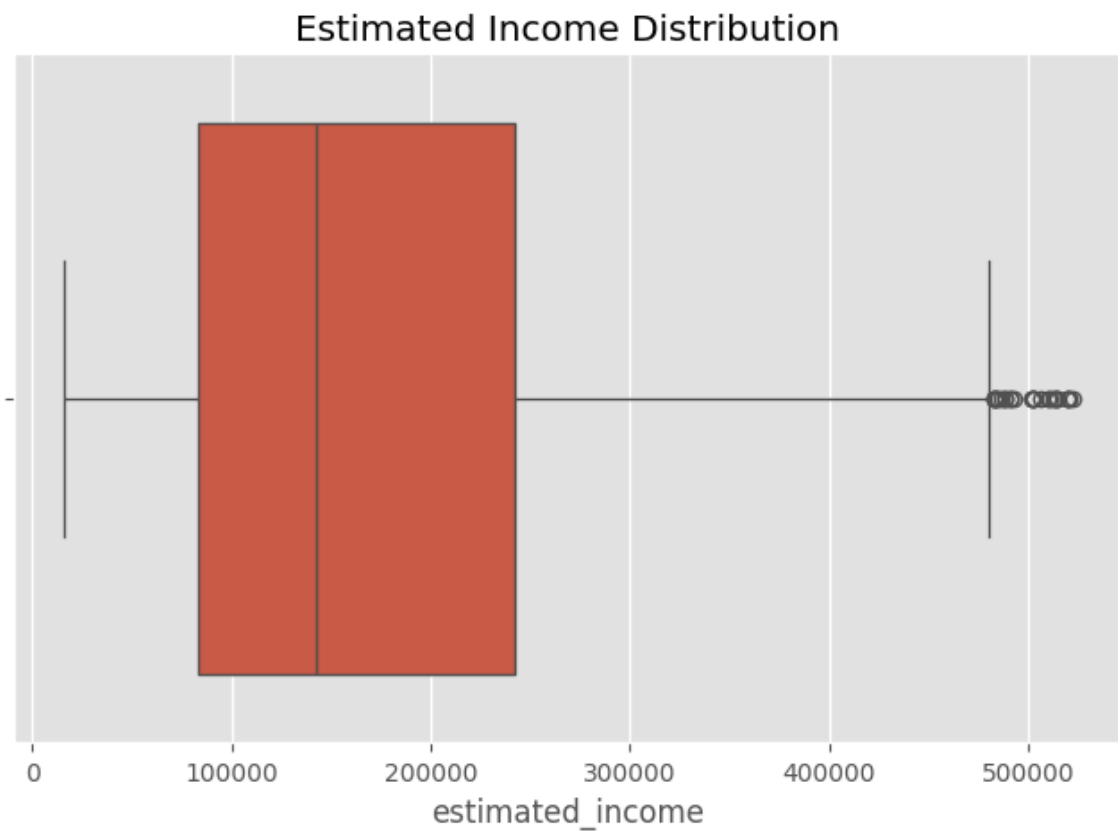


```
plt.figure(figsize=(8,5))

sns.boxplot(x=df["estimated_income"])

plt.title("Estimated Income Distribution")

plt.show()
```



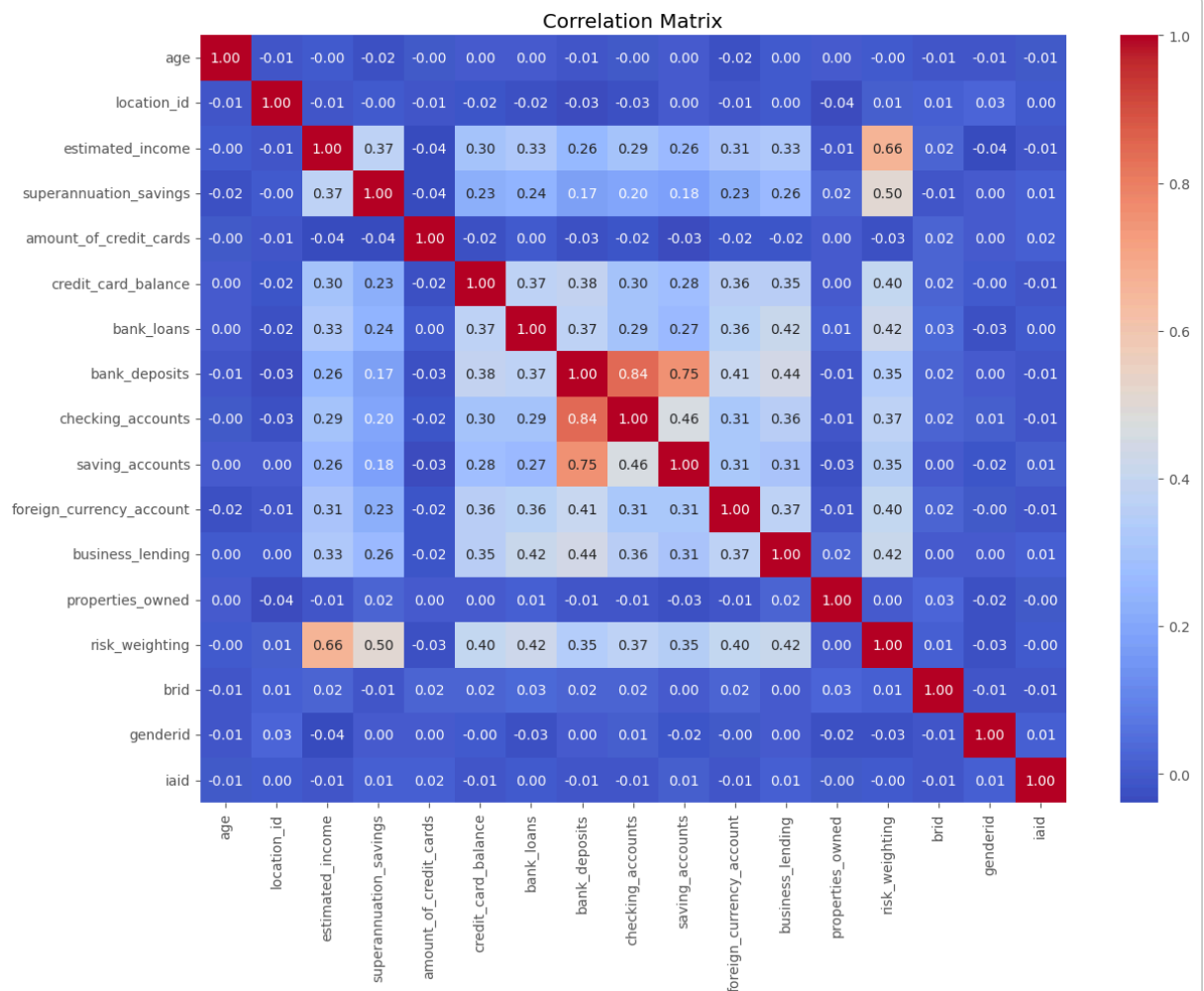
```
plt.figure(figsize=(14,10))

corr = df.select_dtypes(include="number").corr()

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix")

plt.show()
```



```
df.to_csv("clean_banking_data.csv", index=False)
```

```
import os
```

```
os.listdir("/content")
```

```
['.config', 'Banking.xlsx', 'clean_banking_data.csv', 'sample_data']
```

✦ Exploratory Data Analysis

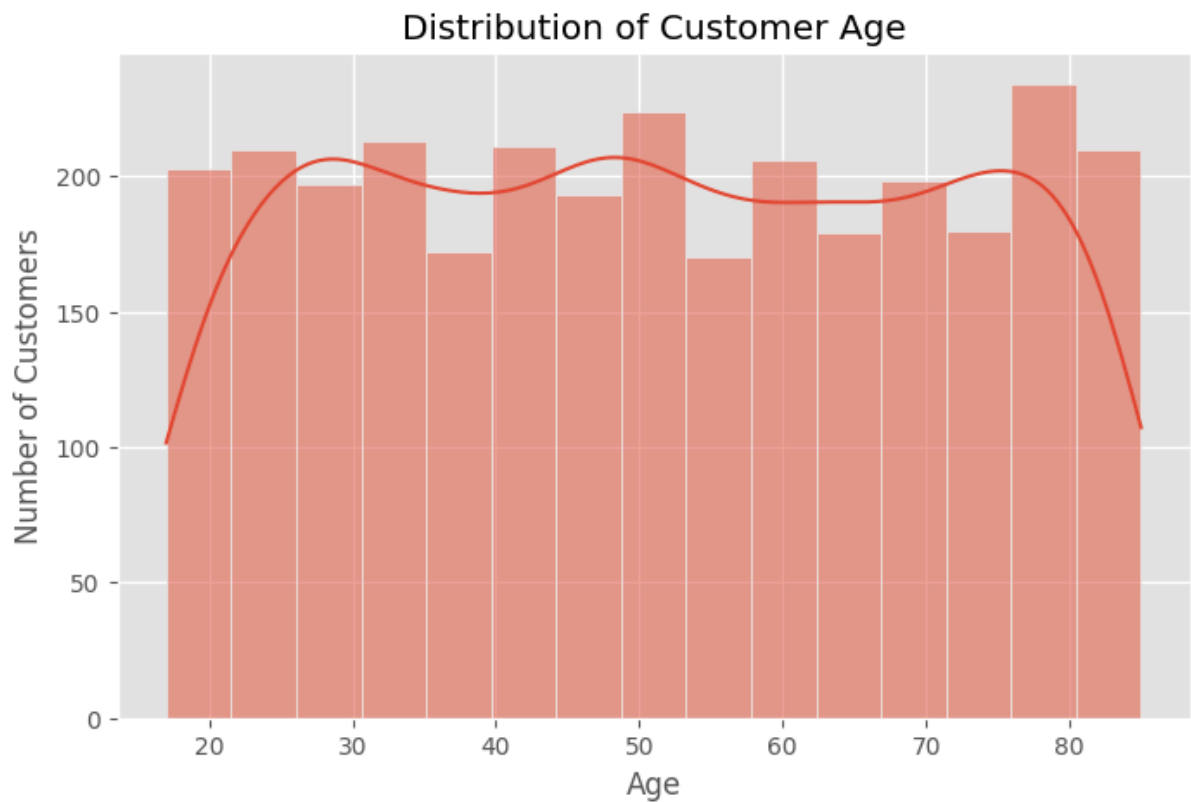
1: What is the age distribution of the bank's customers?

```
plt.figure(figsize=(8,5))

sns.histplot(df["age"], bins=15, kde=True)

plt.title("Distribution of Customer Age")
plt.xlabel("Age")
plt.ylabel("Number of Customers")

plt.show()
```



Most customers belong to the 75 to 80 years age group, indicating that the bank primarily serves old-age peoples.

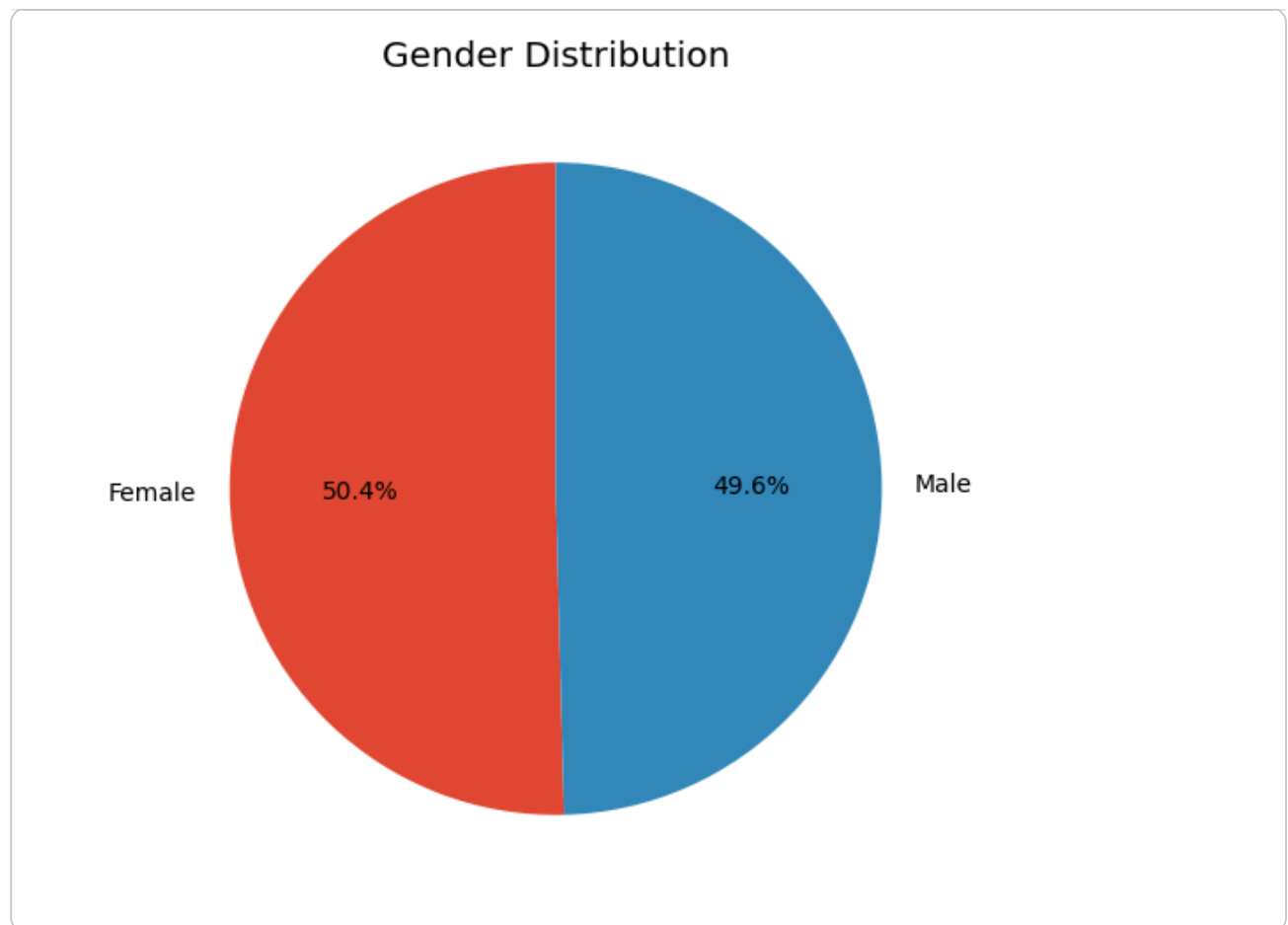
2: Gender Distribution

```
plt.figure(figsize=(6,6))

df["gender"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%",
    startangle=90
)

plt.ylabel("")
plt.title("Gender Distribution")

plt.show()
```



The customer base is fairly balanced between male and female customers.

3: Top Occupations

```
# Top 10 Occupations by Number of Customers

top_10_occupations = (
    df["occupation"]
    .value_counts()
    .head(10)
)

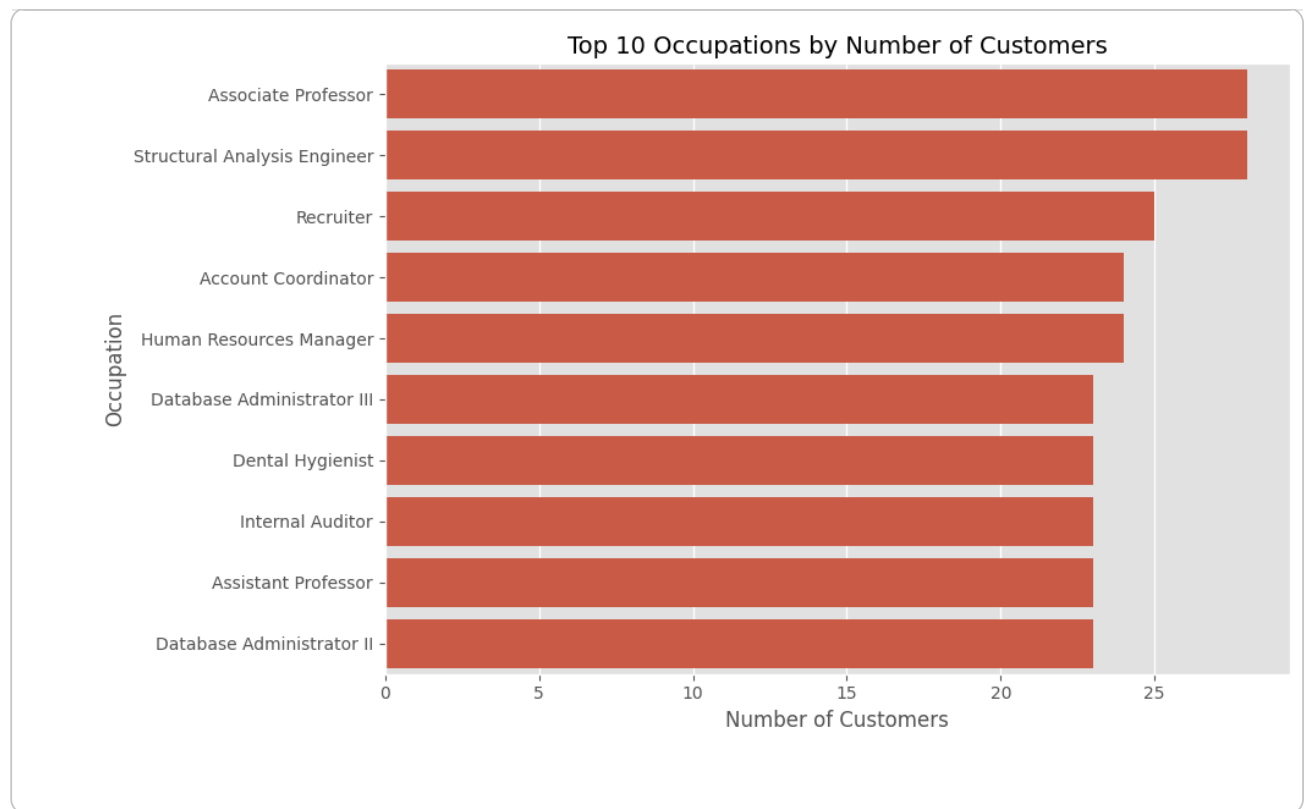
plt.figure(figsize=(10,6))

sns.barplot(
    x=top_10_occupations.values,
    y=top_10_occupations.index
)

plt.title("Top 10 Occupations by Number of Customers", fontsize=14)
plt.xlabel("Number of Customers")
plt.ylabel("Occupation")

plt.tight_layout()

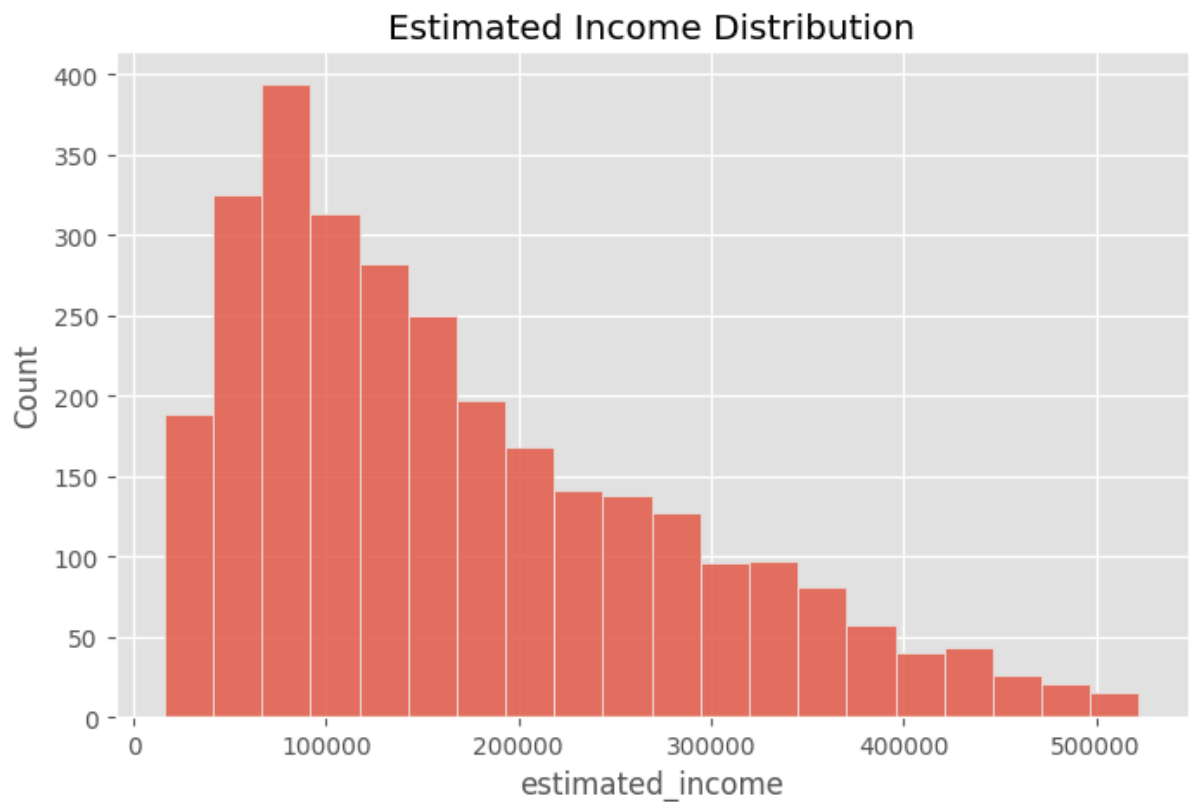
plt.show()
```



The bank has the highest number of customers from Associate Proffesor & Structural Analysis Engineer occupation.

4: Estimated Income Distribution

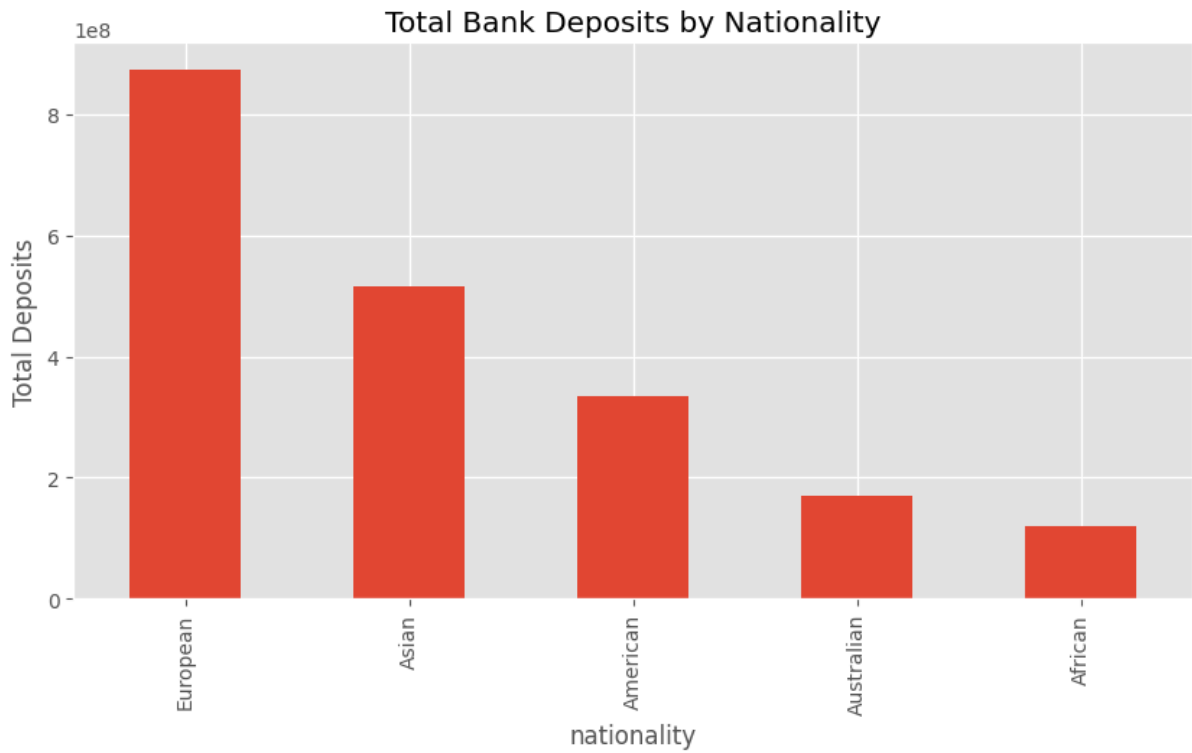
```
plt.figure(figsize=(8,5))  
  
sns.histplot(df["estimated_income"], bins=20)  
  
plt.title("Estimated Income Distribution")  
  
plt.show()
```



Customer income is positively skewed, with a small proportion of high-income customers.

5: Top Nationalities by Deposits

```
deposit_country = (  
    df.groupby("nationality")["bank_deposits"]  
        .sum()  
        .sort_values(ascending=False)  
)  
  
plt.figure(figsize=(10,5))  
  
deposit_country.plot(kind="bar")  
  
plt.title("Total Bank Deposits by Nationality")  
  
plt.ylabel("Total Deposits")  
  
plt.show()
```



Customers from the top nationality contribute the highest total deposits.

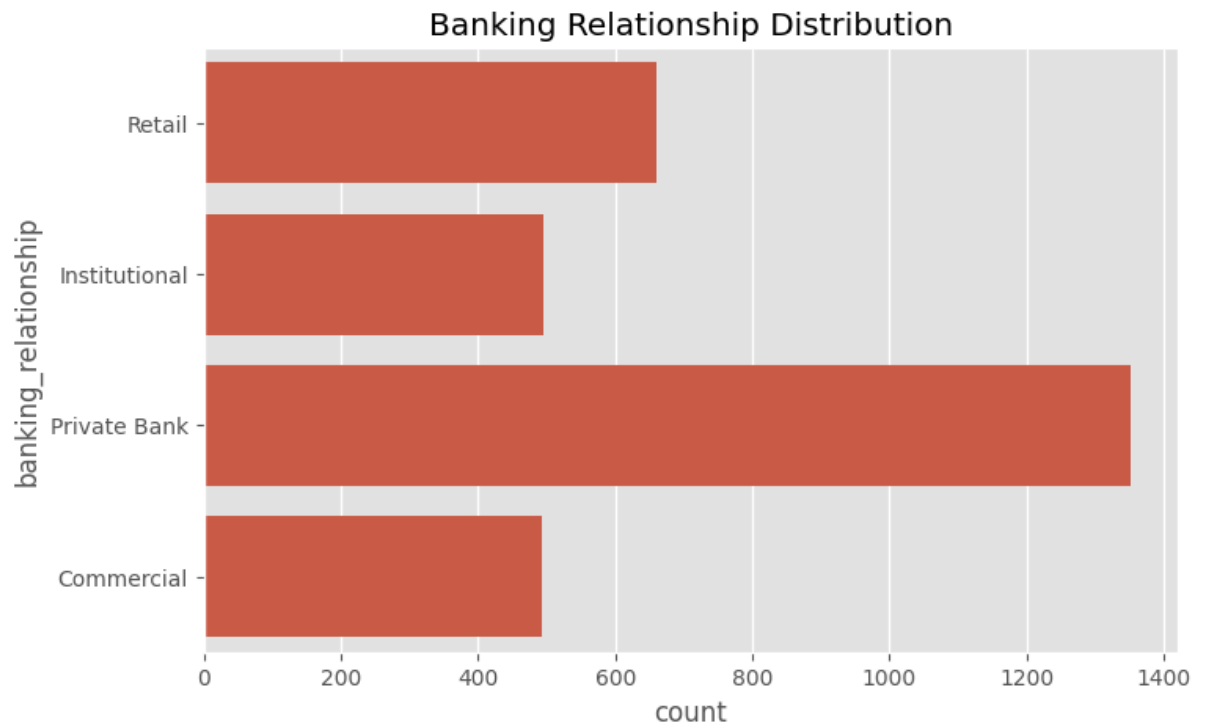
6: Banking Relationship Analysis

```
plt.figure(figsize=(8,5))

sns.countplot(
    data=df,
    y="banking_relationship"
)

plt.title("Banking Relationship Distribution")

plt.show()
```



Most customers fall under the Private banking relationship.

7: Loyalty Classification

```
plt.figure(figsize=(8,5))

sns.countplot(
    data=df,
    x="loyalty_classification"
)

plt.title("Loyalty Classification")

plt.show()
```




The majority of customers belong to the Jade loyalty segment.

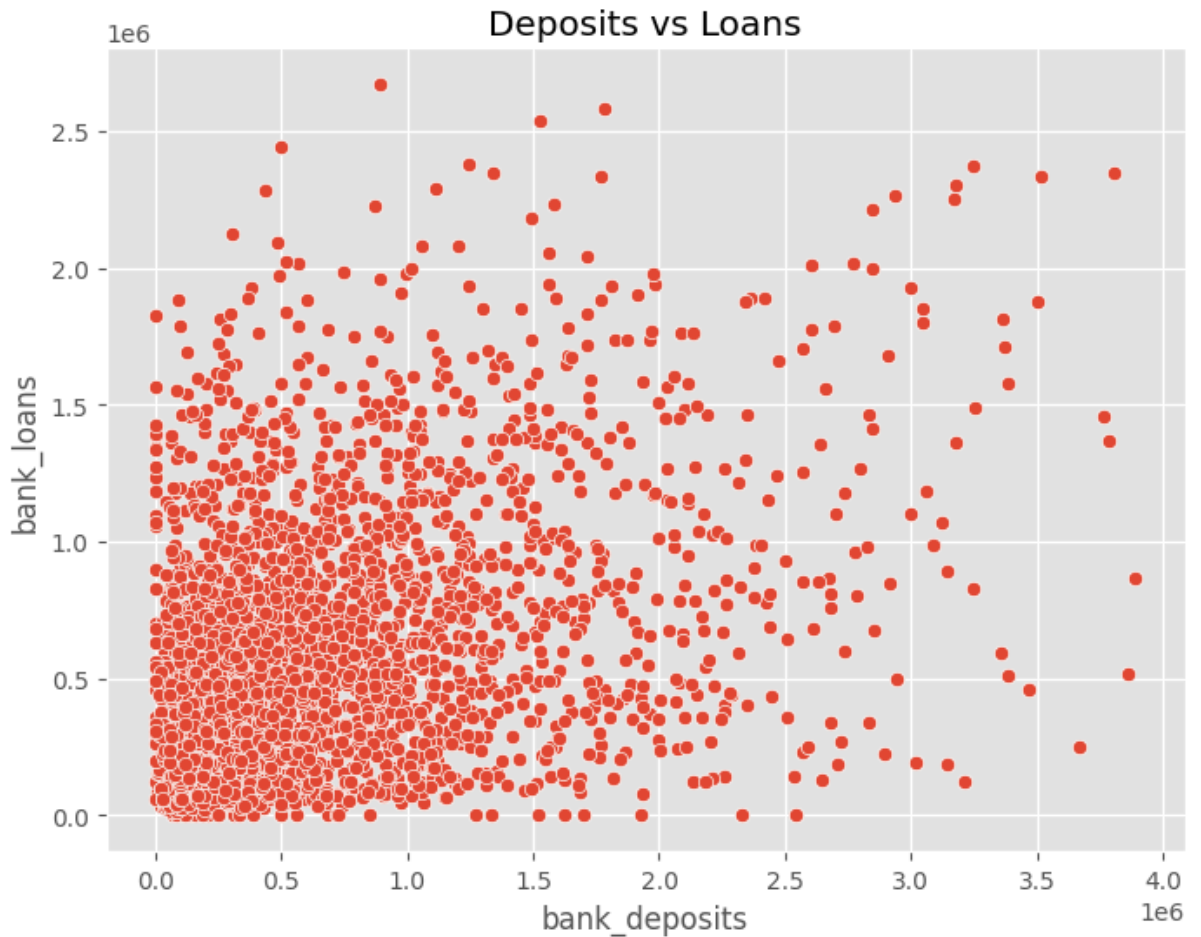
8: Deposits vs Loans

```
plt.figure(figsize=(8,6))

sns.scatterplot(
    data=df,
    x="bank_deposits",
    y="bank_loans"
)

plt.title("Deposits vs Loans")

plt.show()
```



Customers with larger deposits generally tend to maintain higher loan balances.

9: Correlation Heatmap

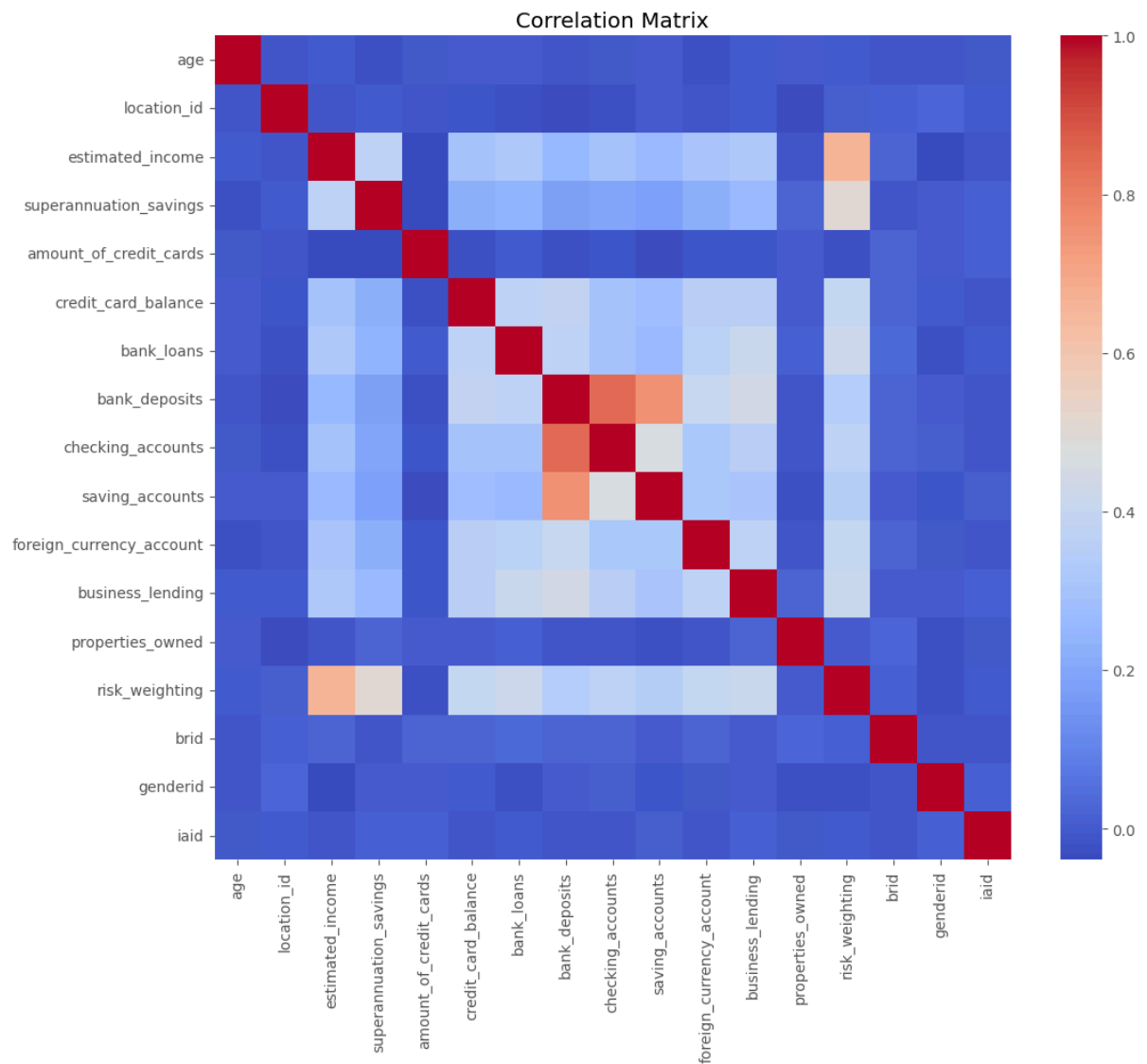
```
plt.figure(figsize=(12,10))

corr = df.select_dtypes(include="number").corr()

sns.heatmap(
    corr,
    cmap="coolwarm",
    annot=False
)

plt.title("Correlation Matrix")

plt.show()
```



Identify the strongest positive and negative relationships between financial variables.

10: Top 10 Customers by Deposits

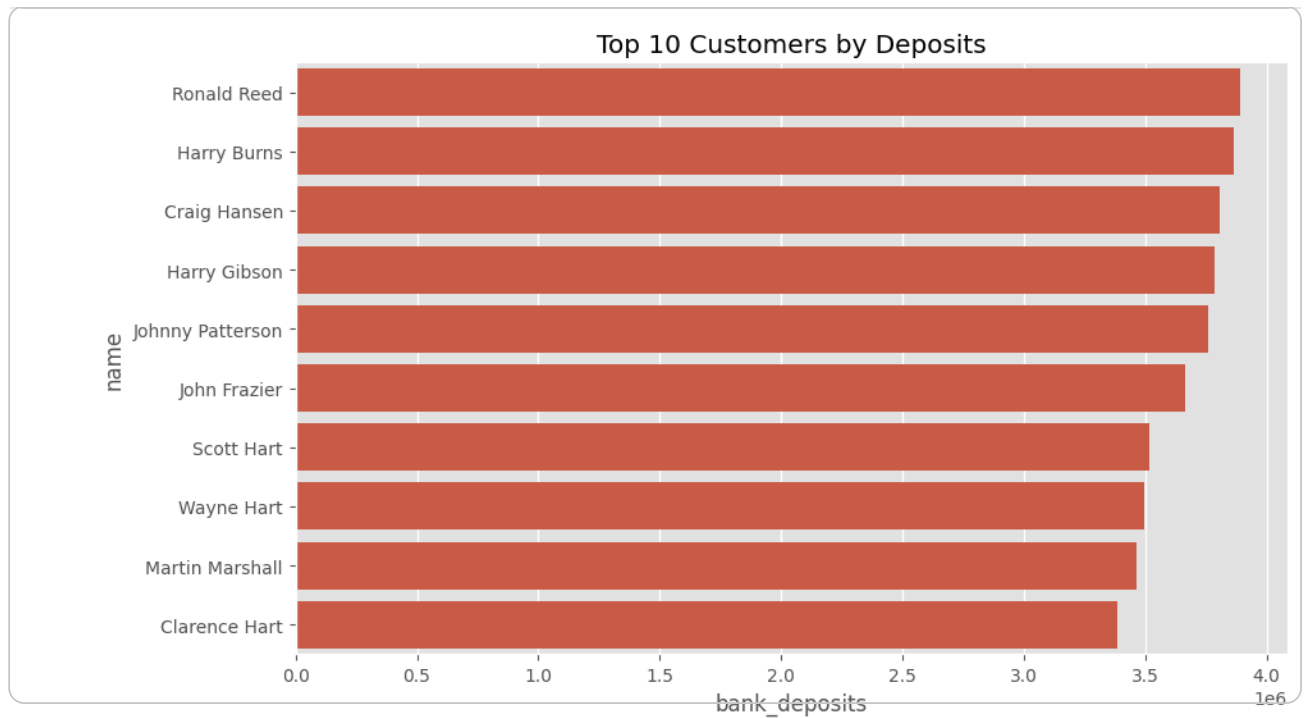
```
top10 = (
    df.nlargest(10, "bank_deposits")
)

plt.figure(figsize=(10,6))

sns.barplot(
    data=top10,
    x="bank_deposits",
    y="name"
)

plt.title("Top 10 Customers by Deposits")

plt.show()
```



These customers represent high-value clients who should receive premium banking services.

11: Risk Weighting Distribution

```
plt.figure(figsize=(8,5))  
  
sns.histplot(df["risk_weighting"], bins=20)  
  
plt.title("Risk Weighting Distribution")  
  
plt.show()
```

